

12.209

AI25BTECH11027 - NAGA BHUVANA

Question:

An Unitary matrix $\begin{pmatrix} ae^{i\alpha} & b \\ ce^{i\beta} & d \end{pmatrix}$ is given where $a, b, c, d, \alpha, \beta$ are real. The inverse of the matrix is

1) $\begin{pmatrix} ae^{i\alpha} & -ce^{i\beta} \\ b & d \end{pmatrix}$ 2) $\begin{pmatrix} ae^{i\alpha} & ce^{i\beta} \\ b & d \end{pmatrix}$ 3) $\begin{pmatrix} ae^{-i\alpha} & b \\ ce^{-i\beta} & d \end{pmatrix}$ 4) $\begin{pmatrix} ae^{-i\alpha} & ce^{-i\beta} \\ b & d \end{pmatrix}$

Solution:

Unitary Matrix: Complex square matrix whose inverse is equal to its conjugate transpose

Let

$$M = \begin{pmatrix} ae^{i\alpha} & b \\ ce^{i\beta} & d \end{pmatrix} \quad (1)$$

$\bar{M}$ be Complex Conjugate Matrix of M

$$\bar{M} = \begin{pmatrix} ae^{-i\alpha} & b \\ ce^{-i\beta} & d \end{pmatrix} \quad (2)$$

$$M^{-1} = \bar{M}^T \quad (3)$$

$$M^{-1} = \begin{pmatrix} ae^{-i\alpha} & b \\ ce^{-i\beta} & d \end{pmatrix}^T \quad (4)$$

$$M^{-1} = \begin{pmatrix} ae^{-i\alpha} & ce^{-i\beta} \\ b & d \end{pmatrix} \quad (5)$$

$\therefore$ Inverse of the Matrix is $\begin{pmatrix} ae^{-i\alpha} & ce^{-i\beta} \\ b & d \end{pmatrix}$